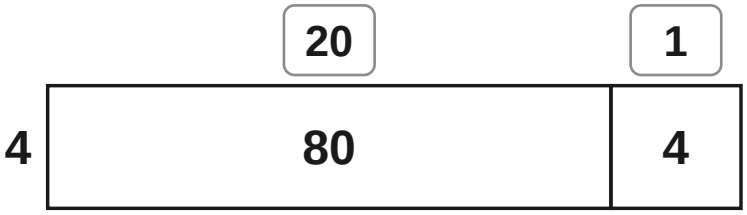
Dividing Bigger Numbers

Split the big number into parts you can divide. Then add.

84 ÷ 4: split 84 into 80 and 4

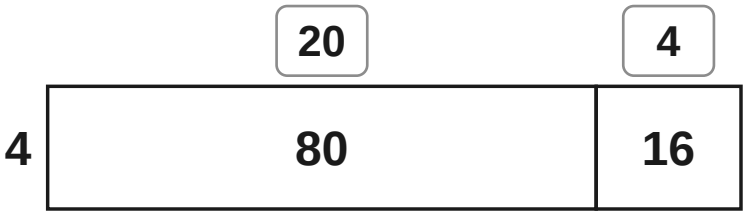


80 ÷ 4 = 20

4 ÷ 4 = 1

20 + 1 = 21

Pick parts that divide easily: 96 ÷ 4



80 is 4 × 20, and 16 is 4 × 4.

20 + 4 = 24

Big numbers: 3,648 ÷ 3

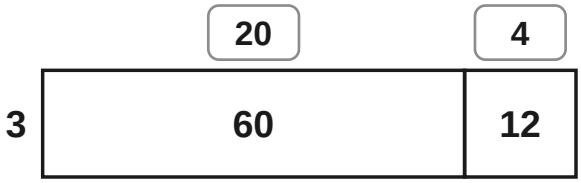
3,000 ÷ 3 = 1,000

600 ÷ 3 = 200

48 ÷ 3 = 16

1,000 + 200 + 16 = 1,216

Your turn!



72 ÷ 3 = 24

For teachers

Goal: divide up to four digits by one digit (4.NBT.6) by splitting the number into parts that divide easily. Say “What part can you share easily? Tens first. Then what's left.”

Dividing Bigger Numbers: Practice

Split the number into easy parts. Divide each one. Add.

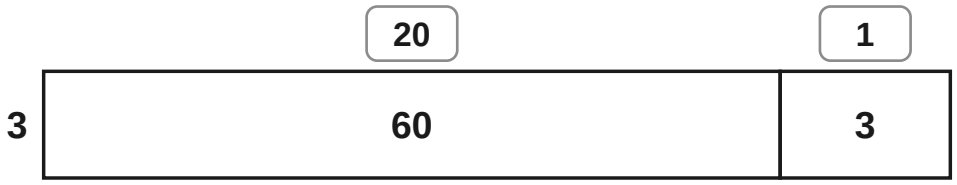
Part 1 Divide tens and hundreds Use a fact you know: 12 ÷ 3 = 4, so 120 ÷ 3 = 40.

1)	80 ÷ 4 = 20	2)	150 ÷ 3 = 50	3)	240 ÷ 6 = 40
4)	90 ÷ 3 = 30	5)	350 ÷ 7 = 50	6)	480 ÷ 8 = 60
7)	60 ÷ 2 = 30	8)	2,400 ÷ 6 = 400	9)	560 ÷ 8 = 70
10)	3,600 ÷ 9 = 400	11)	200 ÷ 5 = 40	12)	1,800 ÷ 2 = 900

Dividing Bigger Numbers: Practice (continued)

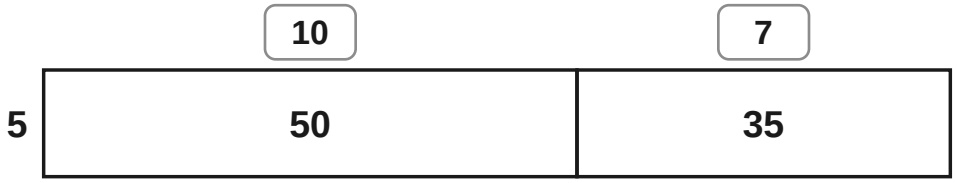
Part 2 The area model The parts are inside. Find the side of each piece.

13) **63 ÷ 3 = 21**



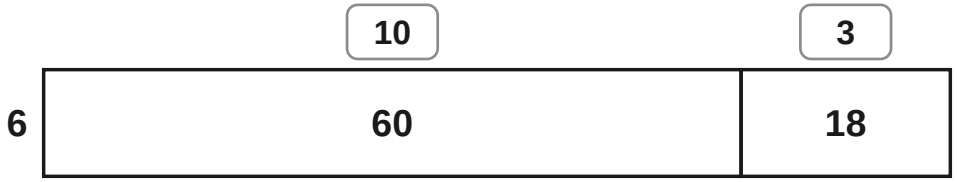
20 + **1** = **21**

14) **85 ÷ 5 = 17**



10 + **7** = **17**

15) **78 ÷ 6 = 13**

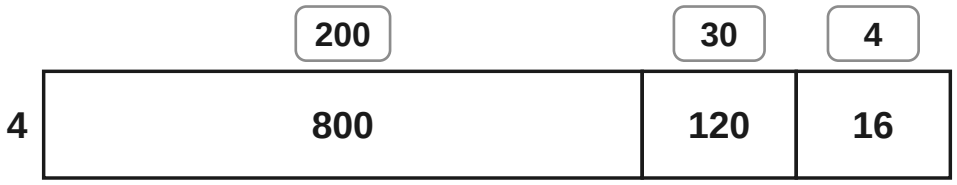


10 + **3** = **13**

Dividing Bigger Numbers: Practice (continued)

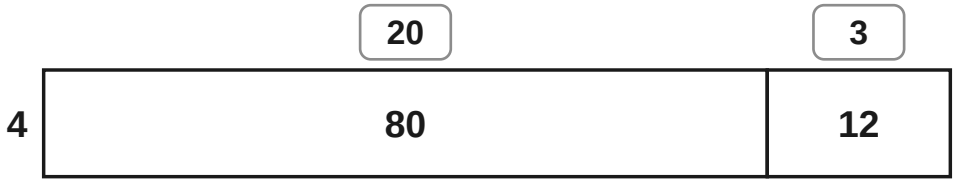
Part 2 The area model (continued)

16) **936 ÷ 4 = 234**



200 + 30 + 4 = 234

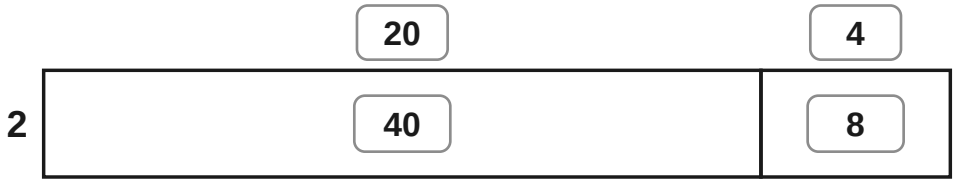
17) **92 ÷ 4 = 23**



20 + 3 = 23

Part 3 Split it yourself Write the parts inside first.

18) **48 ÷ 2 = 24**

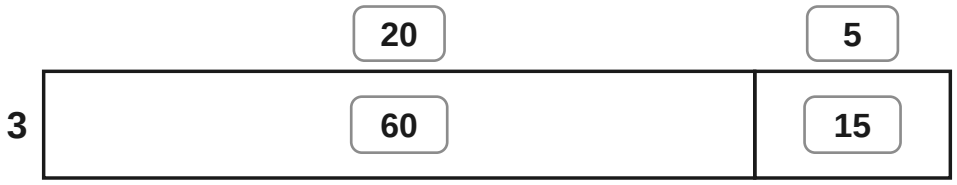


20 + 4 = 24

Dividing Bigger Numbers: Practice (continued)

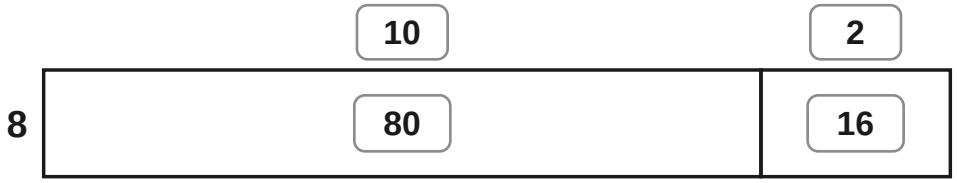
Part 3 Split it yourself (continued)

19) $75 \div 3 = 25$



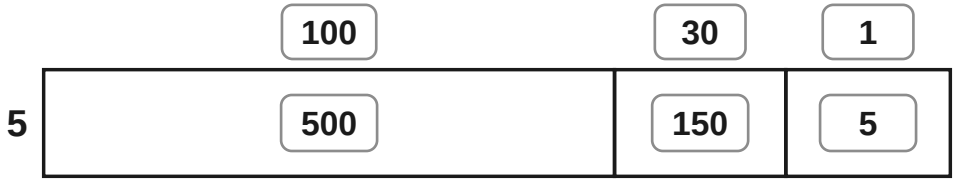
20 + 5 = 25

20) $96 \div 8 = 12$



10 + 2 = 12

21) $655 \div 5 = 131$



100 + 30 + 1 = 131

Dividing Bigger Numbers: Practice (continued)

Part 4 Divide each part No picture this time.

22) $84 = 70 + 14$

$70 \div 7 =$ 10

$14 \div 7 =$ 2

10 + 2 = 12

23) $138 = 120 + 18$

$120 \div 6 =$ 20

$18 \div 6 =$ 3

20 + 3 = 23

24) $464 = 400 + 64$

$400 \div 4 =$ 100

$64 \div 4 =$ 16

100 + 16 = 116

25) $2{,}856 = 2{,}800 + 56$

$2{,}800 \div 4 =$ 700

$56 \div 4 =$ 14

700 + 14 = 714

Part 5 Two digits No help this time.

26) $52 \div 4 =$ 13

27) $87 \div 3 =$ 29

28) $72 \div 6 =$ 12

29) $95 \div 5 =$ 19

30) $68 \div 2 =$ 34

31) $91 \div 7 =$ 13

Dividing Bigger Numbers: Practice (continued)

Part 5 Two digits (continued)

32) $64 \div 4 = 16$	33) $84 \div 6 = 14$	34) $98 \div 7 = 14$
35) $76 \div 2 = 38$	36) $57 \div 3 = 19$	37) $96 \div 6 = 16$

Part 6 Three digits No help this time.

38) $636 \div 6 = 106$	39) $248 \div 2 = 124$	40) $455 \div 5 = 91$
41) $372 \div 3 = 124$	42) $816 \div 8 = 102$	43) $595 \div 7 = 85$

Dividing Bigger Numbers: Practice (continued)

Part 6 Three digits (continued)

44) 432 ÷ 4 = 108

45) 738 ÷ 9 = 82

46) 264 ÷ 2 = 132

Part 7 Four digits No help this time.

47) 4,826 ÷ 2 = 2,413

48) 3,696 ÷ 3 = 1,232

49) 5,025 ÷ 5 = 1,005

50) 8,448 ÷ 4 = 2,112

51) 7,266 ÷ 6 = 1,211

52) 2,835 ÷ 7 = 405

Part 8 With a remainder No help this time. What is left over?

53) 97 ÷ 4 = 24 R 1

54) 85 ÷ 6 = 14 R 1

55) 79 ÷ 3 = 26 R 1

Dividing Bigger Numbers: Practice (continued)

Part 8 With a remainder (continued)

56) 127 ÷ 5 = 25 R 2

57) 98 ÷ 8 = 12 R 2

58) 157 ÷ 7 = 22 R 3

59) 203 ÷ 4 = 50 R 3

60) 89 ÷ 2 = 44 R 1

61) 311 ÷ 6 = 51 R 5

Part 9 Divide

62) 4,823 ÷ 7 = 689

63) 376 ÷ 2 = 188

64) 494 ÷ 2 = 247

65) 2,124 ÷ 4 = 531

66) 1,803 ÷ 3 = 601

67) 870 ÷ 2 = 435

Dividing Bigger Numbers: Practice (continued)

Part 9 Divide (continued)

68) 3,880 ÷ 5 = 776

69) 2,412 ÷ 6 = 402

70) 3,135 ÷ 5 = 627

71) 2,100 ÷ 4 = 525

72) 447 ÷ 3 = 149

73) 2,295 ÷ 9 = 255

74) 282 ÷ 6 = 47

75) 502 ÷ 2 = 251

76) 2,975 ÷ 7 = 425

77) 446 ÷ 2 = 223

78) 306 ÷ 6 = 51

79) 2,184 ÷ 4 = 546

Dividing Bigger Numbers: Practice (continued)

Part 9 Divide (continued)

80) 1,944 ÷ 8 = 243

81) 8,343 ÷ 9 = 927

82) 1,136 ÷ 2 = 568

83) 6,498 ÷ 9 = 722

84) 1,532 ÷ 2 = 766

85) 504 ÷ 2 = 252

86) 3,804 ÷ 6 = 634

87) 5,504 ÷ 8 = 688

88) 752 ÷ 8 = 94

89) 4,425 ÷ 5 = 885

90) 1,211 ÷ 7 = 173

91) 1,864 ÷ 4 = 466

Dividing Bigger Numbers: Practice (continued)

Part 10 What is left?

92) **147 ÷ 5 =** 29 **R** 2

93) **835 ÷ 8 =** 104 **R** 3

94) **515 ÷ 3 =** 171 **R** 2

95) **286 ÷ 6 =** 47 **R** 4

96) **476 ÷ 3 =** 158 **R** 2

97) **664 ÷ 6 =** 110 **R** 4

98) **776 ÷ 5 =** 155 **R** 1

99) **725 ÷ 4 =** 181 **R** 1

100) **398 ÷ 3 =** 132 **R** 2

101) **211 ÷ 5 =** 42 **R** 1

102) **591 ÷ 9 =** 65 **R** 6

103) **577 ÷ 7 =** 82 **R** 3

Dividing Bigger Numbers: Practice (continued)

Part 10 What is left? (continued)

104) 245 ÷ 3 = 81 R 2

105) 161 ÷ 6 = 26 R 5

106) 298 ÷ 8 = 37 R 2

Part 11 Remember? Problems from earlier packets.

107) 26 × 14 = 364

108) 40 × 70 = 2,800

109) 5,729 to the nearest hundred = 5,700

110) $\frac{2}{5}$ of 30 = 12

111) 6,000 − 2,385 = 3,615

112) $\frac{14}{4}$ = 3 wholes and $\frac{2}{4}$

Part 12 Word problems Share it, or make groups.

113) 96 stickers are shared equally by 4 friends. How many stickers does each friend get?

Number sentence 96 ÷ 4 = 24

Answer 24 stickers

Draw or show your work

114) A farmer packs 252 eggs into cartons of 6. How many cartons does she fill?

Number sentence 252 ÷ 6 = 42

Answer 42 cartons

Draw or show your work

115) A school has 1,575 books to put on 5 shelves equally. How many books go on each shelf?

Number sentence 1,575 ÷ 5 = 315

Answer 315 books

Draw or show your work

Dividing Bigger Numbers: Practice (continued)

Part 12 Word problems (continued)

116) 98 kids ride in vans that hold 8 kids each. How many vans are full? How many kids are left?

Number sentence **98 ÷ 8 = 12 R 2**

Answer **12 R 2**

Draw or show your work

117) Challenge: 135 kids go on a trip. Each van holds 8 kids. How many vans do they need for everyone?

Number sentence **135 ÷ 8 = 16 R 7, so 17 vans**

Answer **17** vans

Draw or show your work

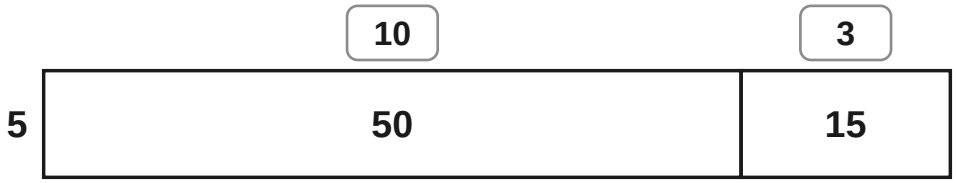
Check-In: Dividing Bigger Numbers

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 The area model

1) **65 ÷ 5 = 13**



10 + 3 = 13

Part 2 Two and three digits

2) **542 ÷ 2 = 271**

3) **835 ÷ 5 = 167**

4) **412 ÷ 2 = 206**

Part 3 Four digits

5) **4,052 ÷ 4 = 1,013**

6) **2,026 ÷ 2 = 1,013**

Part 4 With a remainder

7) **394 ÷ 8 = 49 R 2**

8) **357 ÷ 9 = 39 R 6**

9) **172 ÷ 5 = 34 R 2**